CIS 5190: Homework 3

Written Answers

Use of AI Tools

I used AI for concepts learning as well as answer checks. I also used it to transform my written answer to Latex format. Specifically, I used OpenAI Codex, which also generated a draft of the solutions and figures in this document. The explanations can help me follow the steps, but I still need to check the reasoning myself so that I do not skip understanding the material.

Time Spent

The following are AI-suggested estimates, not measured times. They should be checked against my actual time before submission.

Part	Estimated hours
Reading the assignment	0.25
Question 1	0.50
Question 2	0.50
Question 3	0.75
Question 4	0.25
Question 5	0.50
Total	2.75

Question 1: K-means

(a) Two iterations

For a point (x, y) and centroid (a, b), the Euclidean distance is

$$d = \sqrt{(x-a)^2 + (y-b)^2}.$$

We assign each point to its nearest centroid, then average the points in each cluster.

Iteration 1. Start with $\mu_1 = (6, 9)$ and $\mu_2 = (8, 4)$.

Sample	$d(S, 1)$	$d(S, 2)$	Assigned cluster
$A = (2, 3)$	$\sqrt{52} = 2\sqrt{13}$	$\sqrt{37}$	2
$B = (4, 6)$	$\sqrt{13}$	$\sqrt{20} = 2\sqrt{5}$	1
$C = (5, 1)$	$\sqrt{65}$	$\sqrt{10}$	2
$D = (10, 12)$	$\sqrt{25} = 5$	$\sqrt{68} = 2\sqrt{17}$	1

The new clusters and centroids are

$$C_1 = \{B, D\}, \qquad \mu_1 = \left(\frac{4+10}{2}, \frac{6+12}{2}\right) = (7, 9),$$
$$C_2 = \{A, C\}, \qquad \mu_2 = \left(\frac{2+5}{2}, \frac{3+1}{2}\right) = \left(\frac{7}{2}, 2\right).$$

Iteration 2. Now use $\mu_1 = (7, 9)$ and $\mu_2 = (7/2, 2)$.

Sample	$d(S, 1)$	$d(S, 2)$	Assigned cluster
A	$\sqrt{61}$	$\sqrt{13}/2$	2
B	$\sqrt{18} = 3\sqrt{2}$	$\sqrt{65}/2$	2
C	$\sqrt{73}$	$\sqrt{13}/2$	2
D	$\sqrt{18} = 3\sqrt{2}$	$\sqrt{569}/2$	1

The new clusters and centroids are

$$C_1 = \{D\}, \qquad \mu_1 = (10, 12),$$
$$C_2 = \{A, B, C\}, \qquad \mu_2 = \left(\frac{2+4+5}{3}, \frac{3+6+1}{3}\right) = \left(\frac{11}{3}, \frac{10}{3}\right).$$

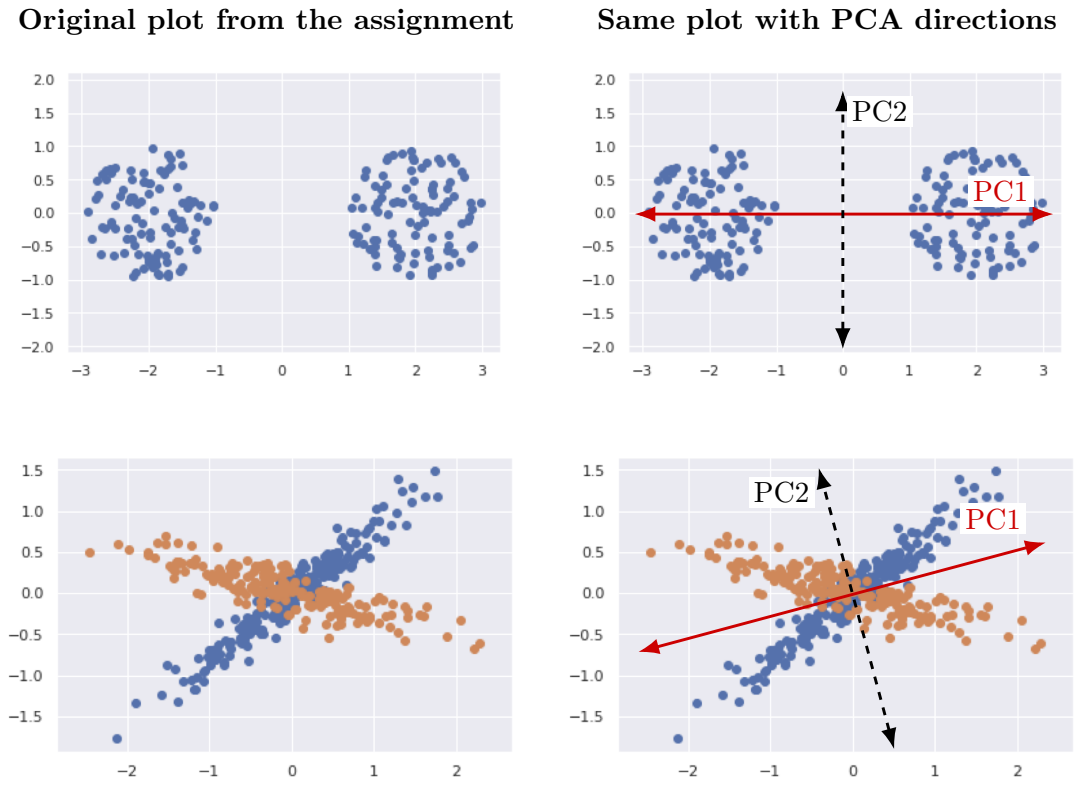
(b) Four clusters

The four centroids are $\boxed{(2, 3),\ (4, 6),\ (5, 1),\ (10, 12)}$, in any order. Each point has its own centroid, so the total squared distance is zero. This is the global minimum because a squared distance cannot be negative.

Question 2: PCA

(a) Principal-component directions

Both principal-component lines pass through the overall data mean. PC1 follows the direction with the largest spread, and PC2 is perpendicular to PC1. The colors do not define separate PCA problems.



Red solid line: PC1. Black dashed line: PC2. The lines cross at the approximate overall mean.

Left: The greatest spread is between the two groups, so PC1 is approximately horizontal and PC2 is approximately vertical.

Right: Looking at all points together, the spread is mostly horizontal with a small upward tilt. PC1 therefore has a shallow positive slope, and PC2 has a steep negative slope. These are visual estimates; the exact angles cannot be read from the plot. Reversing either direction gives the same principal component.

(b) Retaining at least 75% of the variance

The total variance is

$$2.1 + 1.8 + 1.3 + 0.9 + 0.4 + 0.2 + 0.15 + 0.02 + 0.001 = 6.871.$$

The required amount is $0.75(6.871) = 5.15325$. The first two eigenvalues sum to $3.9 < 5.15325$, while the first three sum to $5.2 > 5.15325$. Therefore, $\boxed{K = 3}$, retaining approximately 75.68% of the variance.

Question 3: PCA by Hand

(a) Unit-vector principal components

The mean is

$$\bar{x} = \left(\frac{4+2+5+1}{4}, \frac{1+3+4+0}{4}\right) = (3, 2).$$

Subtract the mean from each row:

$$Z = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 2 & 2 \\ -2 & -2 \end{bmatrix}.$$

Using the covariance convention $\Sigma = Z^\top Z/4$,

$$\Sigma = \frac{1}{4}\begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix} = \begin{bmatrix} 5/2 & 3/2 \\ 3/2 & 5/2 \end{bmatrix}.$$

Its eigenvalues satisfy

$$\det(\Sigma - \lambda I) = \left(\frac{5}{2} - \lambda\right)^2 - \left(\frac{3}{2}\right)^2 = (\lambda - 4)(\lambda - 1) = 0.$$

For $\lambda_1 = 4$, the eigenvector satisfies $v_x = v_y$. For $\lambda_2 = 1$, it satisfies $v_x = -v_y$. Normalizing gives

$$\boxed{v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}}, \qquad \boxed{v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}}.$$

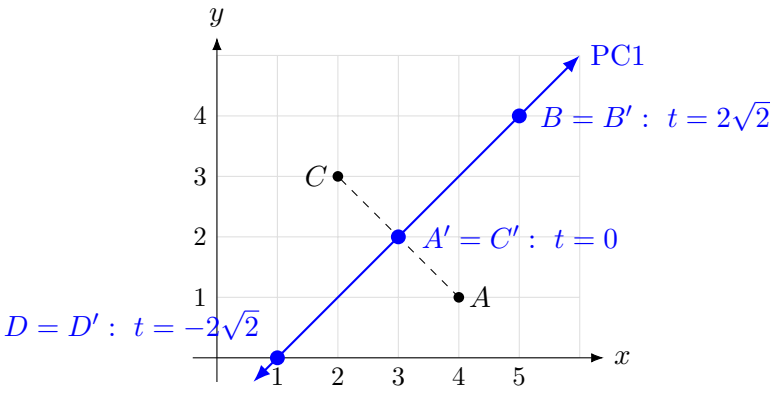
Bob should use v_1, since it retains the largest variance: $4/(4+1) = 80\%$. Using $1/(n-1)$ instead of $1/n$ changes the eigenvalues but not these directions or the retained fraction.

(b) Centered projections

For each point, its one-dimensional coordinate is $t = v_1^\top(x - \bar{x})$, and its projected point on the original plot is $\hat{x} = \bar{x} + tv_1$. The PC1 line passes through $(3, 2)$ and has equation $y = x - 1$.

Point	Centered point	Coordinate t	Projected point $\hat{x}$
$A = (4, 1)$	$(1, -1)$	0	$(3, 2)$
$B = (5, 4)$	$(2, 2)$	$2\sqrt{2}$	$(5, 4)$
$C = (2, 3)$	$(-1, 1)$	0	$(3, 2)$
$D = (1, 0)$	$(-2, -2)$	$-2\sqrt{2}$	$(1, 0)$

In the original matrix row order, Bob sends $\boxed{(0,\ 0,\ 2\sqrt{2},\ -2\sqrt{2})}$.


y
x
PC1
$B = B' : \ t = 2\sqrt{2}$
$A' = C' : \ t = 0$
$D = D' : \ t = -2\sqrt{2}$
C
A
1
2
3
4
5

Question 4: Image Filtering / Convolution

(a) Mean filter: image B

$$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

Each output pixel is the average of a 3×3 neighborhood. Averaging smooths small details and blurs the image, so the answer is $\boxed{\text{B}}$.

(b) Doubling pixel values: image A

$$\begin{bmatrix}0 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 0\end{bmatrix}$$

The displayed output is $\min(2p, 1)$ for an original pixel value $p \in [0, 1]$. This brightens the image and clips values of at least 0.5 to white, so the answer is $\boxed{\text{A}}$.

(c) Horizontal intensity differences: image C

$$\begin{bmatrix}1 & 0 & -1\\ 2 & 0 & -2\\ 1 & 0 & -1\end{bmatrix}$$

This filter compares the left and right sides of a neighborhood, so it detects changes across the horizontal direction and emphasizes vertical edges. Image $\boxed{\text{C}}$ shows those edges, including the vertical tree trunks.

Question 5: CNN Dimensions and Parameters

For a convolution or pooling layer, each spatial output size is

$$n_{\text{out}} = \left\lfloor \frac{n_{\text{in}} + 2p - k}{s} \right\rfloor + 1.$$

The padding preserves the height and width in each convolution. Each pooling layer halves both dimensions, and ReLU does not change the shape.

Layer	Output shape (height × width × channels)
Input	$320 \times 320 \times 3$
Convolution 1 + ReLU	$320 \times 320 \times 16$
Max pooling 1	$160 \times 160 \times 16$
Convolution 2 + ReLU	$160 \times 160 \times 8$
Max pooling 2	$80 \times 80 \times 8$
Convolution 3 + ReLU	$80 \times 80 \times 4$
Max pooling 3	$40 \times 40 \times 4$

(a) Output height

$\boxed{40}$ pixels, since $320/2^3 = 40$.

(b) Output width

$\boxed{40}$ pixels, by the same calculation.

(c) Output channels

$\boxed{4}$, the number of output channels in the last convolution.

(d) Learnable parameters

For a convolutional layer, the count is

$$(k^2 C_{\text{in}} + 1) C_{\text{out}}.$$

Each output channel has $k^2 C_{\text{in}}$ weights and one bias.

$$\begin{aligned} &\text{Convolution 1:} && (5 \cdot 5 \cdot 3 + 1) \cdot 16 = 1216, \\ &\text{Convolution 2:} && (3 \cdot 3 \cdot 16 + 1) \cdot 8 = 1160, \\ &\text{Convolution 3:} && (3 \cdot 3 \cdot 8 + 1) \cdot 4 = 292. \end{aligned}$$

ReLU and max pooling have no learnable parameters, so the total is

$$\boxed{1216 + 1160 + 292 = 2668}.$$